

What Aggregate Accuracy Hides: A Powered Subject-Level Audit of Prompt Optimization

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Abstract

Automatic prompt optimizers are marketed on aggregate accuracy and increasingly applied to low-resource languages. On a subject-partitioned Uzbek university-entrance benchmark we run the audit that aggregate accuracy prevents: with 5,058 paired items from a frozen corpus never used in development, DSPY’s BOOTSTRAPFEWSHOT improves reasoning subjects (+2.3, $p=0.029$) and does not improve knowledge subjects (−1.2, $p=0.15$), leaving overall accuracy flat (+0.1, $p=0.92$). The reasoning gain and the knowledge shortfall offset each other, so a practitioner reading one number sees nothing at all. Tested directly rather than as two marginal tests, the reallocation is significant across items (knowledge-versus-reasoning odds ratio 1.40 [1.08, 1.81]) but not across models, which do not share a common effect ($I^2=66\%$; treating models as exchangeable gives 1.33 [0.83, 2.14]). The users’ own language is simply a knowledge subject: it is not distinguishable from the benchmark’s other knowledge subject, which fares marginally worse (−1.1 vs. −1.4). We report this against our own initial finding, which the same data overturns. At $n\approx 100$ per model—the scale at which subject-level audits are normally run—we measured a native-language-specific effect ($p=0.011$; odds ratio 2.27 against non-native items, with the benchmark’s other knowledge subject at a clean 1.04). Powered on held-out data it dissolves: the odds ratio falls to 1.25 [0.98, 1.61] and the other knowledge subject rises to 1.16, statistically indistinguishable. A second correction follows the same pattern: a one-line brevity-constraint repair we proposed turns out to be inert ($p=0.90$). Substituting different correct demonstrations does help ($p=0.0097$), which would indict correctness-based selection itself—but we show that a fixed shuffle seed makes six models select only eight distinct demonstrations between them, so that contrast rests on one BOOTSTRAPFEWSHOT

draw and we report it as an observation rather than a finding. We release the harness, per-item logs, and a one-command verifier for every number.

1 Introduction

Chain-of-thought prompting (Wei et al., 2022) helps mainly on math and symbolic reasoning (Sprague et al., 2025) and can hurt elsewhere (Liu et al., 2025). Layered on top, automatic prompt optimizers—most prominently DSPY (Khatab et al., 2024) with its BOOTSTRAPFEWSHOT teleprompter, which retains as few-shot demonstrations only training examples the model itself answers correctly—are sold as turnkey accuracy boosters and increasingly applied to low-resource languages.

They are also evaluated the way they are sold: on one aggregate number. This paper asks what that number conceals, on a benchmark whose subject partition makes the question answerable, and at a sample size that makes the answer stable.

The finding. BOOTSTRAPFEWSHOT does not raise accuracy uniformly. It raises *reasoning*-subject accuracy and leaves *knowledge*-subject accuracy where it was, and on our benchmark the two move the aggregate by a tenth of a point between them (§4). Aggregate accuracy is not merely a coarse summary here; it is blind by construction. Any deployment whose users care disproportionately about the knowledge half—which, for a national-language application, is the whole point—is paying for a gain it does not receive.

The correction. Our own first pass through this data reached a sharper and more alarming conclusion: that the harm was specific to the *native language*. At $n\approx 100$ items per model, that finding looked solid—a significant pooled effect, a 2.27 odds ratio against non-native items, and a placebo

rotation in which the benchmark’s other knowledge subject sat at 1.04. Re-run on a frozen corpus at four times the scale, it dissolves (§4). The native subject and the other knowledge subject become indistinguishable. We report both, because the gap between them is the most useful thing we found: subject-level audits at the scale they are normally run at can manufacture a language-specific story out of a subject-type effect.

A second correction, same shape. We proposed a one-line repair—require bootstrapped demonstrations to satisfy the prompt’s own brevity instruction—and measured it recovering most of the loss. An ablation separating the two things that repair changes at once shows the constraint itself is inert ($p=0.90$); what helps is substituting different demonstrations. That would indict correctness-based selection, except that the optimizer’s selection turns out to be near-degenerate: a fixed pool ordering leads six models to choose only eight distinct demonstrations between them, so the contrast compares one BOOTSTRAPFEWSHOT draw against several random ones (§5). We report it as an observation, name the experiment that would settle it, and do not build on it.

What survives unchanged. A format route from optimizer to failure is real and causally verified: correctness-selection imports verbose demonstrations, a fixed decoding budget truncates them, and raising the budget improves accuracy in exactly the models that truncate and not in a zero-truncation control (§6). It accounts for 17.6% of harmful flips at scale—a genuine and repairable route, and a minority of the effect.

2 Related Work

CoT and prompt optimization. CoT’s gains concentrate on math and symbolic tasks (Sprague et al., 2025); it can reduce accuracy where deliberation hurts humans (Liu et al., 2025), and its marginal value on strong models is shrinking (Meincke et al., 2025). DSPY (Khatab et al., 2024) and MIPROV2 (Opsahl-Ong et al., 2024) self-bootstrap demonstrations. Sarmah et al. (2024) conjectured that teleprompters over-rely on examples the model already answers well. We find that skew is an artifact of comparing against the wrong baseline (§6): what the selected demonstrations share is verbosity, not subject.

Format restrictions and budgets. Format constraints degrade reasoning performance (Tam et al., 2024), and tokenizers price languages unequally (Petrov et al., 2023) with accuracy consequences across low-resource languages (Lundin et al., 2025). We connect this to prompt optimization: the optimizer *generates* format pressure, which fixed budgets convert into failures—though through input length and induced verbosity rather than output-side fertility (§7).

Cross-lingual reasoning dynamics. A parallel literature asks which language multilingual models reason *in* (Wendler et al., 2024; Schut et al., 2025; Zhong et al., 2024), and how language-centric pretraining shapes reasoning (Park et al., 2025; Tran et al., 2025). Our powered result reduces the pull of these accounts for this phenomenon: what we observe separates subject *types*, not languages.

Disparate impact and evaluation scale. Zhu et al. (2022) show semi-supervised learning can help high-baseline subgroups at low-baseline subgroups’ expense, masked by averages. Our result is the prompt-optimization analogue, with the subgroup axis turning out to be subject type. Native Turkic benchmarks include TUMLU (Isbarov et al., 2025), KazMMLU (Togmanov et al., 2025) and TurkishMMLU (Yüksel et al., 2024); DTM (Hazratov et al., 2026) provides the subject partition our analyses require.

3 Setup

Data. DTM (Hazratov et al., 2026): 1,000 four-option Uzbek questions (ONA_TILI, Uzbek grammar and literature, 400; TARIX, history, 300; mathematics 150; physics 150). Seeded stratified split (train 500/dev 249/test 251). Eight rows of the release file carry nulled answers: two fall in the test split and were recovered from run logs, six are excluded. **Replication corpus:** the DTM 2019 public release contributes 2,028 deduplicated items across the same four subjects (393 ONA_TILI, 504 TARIX, 727 mathematics, 404 physics), never used in any development decision, with its protocol frozen before first model contact. This is the evaluation set for §4.

Models and engines. Six open instruction-tuned LLMs spanning GEMMA-4 (E4B, 31B) and QWEN-3.5/3.6 (4–27B), served with Ollama at 4-bit (Q4_K_M) on an NVIDIA DGX Spark, plus

Table 1: The powered audit: CoT→BOOTSTRAP on the frozen replication corpus, six models, 5,058 paired items. Reasoning subjects gain significantly; knowledge subjects do not; the aggregate is flat to a tenth of a point. The native subject is not distinguishable from the benchmark’s other knowledge subject.

Subject	CoT	BOOTSTRAP	Δ	n	p
ONA_TILI (<i>native</i>)	35.8	34.8	−1.1	2358	.283
TARIX (<i>history</i>)	55.9	54.4	−1.4	900	.344
physics	69.6	71.1	+1.6	900	—
mathematics	69.9	73.0	+3.1	900	—
knowledge (both)	41.4	40.2	−1.2	3258	.150
reasoning	69.7	72.1	+2.3	1800	.029
overall	51.5	51.5	+0.1	5058	.925

vLLM at bf16 for the three large models and an Apple-silicon replication. Greedy decoding, max_tokens 512 unless manipulated.

Conditions. **cot** (brevity-constrained zero-shot CoT: “at most two sentences”), **bootstrap** (BOOTSTRAPFEWSHOT on the CoT module, 4 demonstrations, correctness metric), **bootstrap-compliant** (the metric additionally requires the demonstration to satisfy the brevity instruction), and **random- k** (four demonstrations drawn uniformly from the model’s own correct traces, §5).

Instrumentation. Every generation logs raw completion text, token counts, finish reason and a parse-failure flag. We score two ways: *deployment* (a parse failure is wrong—what a user experiences) and *knowledge* (items where either condition truncated or failed to parse are excluded).

Statistics. Paired exact binomial McNemar throughout, Holm-corrected within each contrast family. Subject contrasts use a Mantel–Haenszel odds ratio stratified by model, with intervals from a bootstrap resampling *items* as whole clusters, since all models answer the same items. Every number regenerates from committed per-item logs by a single command in the released harness.

4 What the Aggregate Hides

We evaluated the demonstrations BOOTSTRAPFEWSHOT selects—the identical compiled programs, not a fresh compile—against the frozen four-subject replication corpus, for all six models (Table 1). Three things follow.

The optimizer works, on half the benchmark. Reasoning subjects gain +2.3 points ($b=155$,

$c=197$, $n=1800$, $p=0.029$). That is the effect BOOTSTRAPFEWSHOT is sold on and it is real.

Knowledge subjects do not benefit. Pooled, they move −1.2 ($b=349$, $c=311$, $p=0.150$)—directionally negative, not significant. We state the conservative form: the optimizer does not *help* knowledge subjects. Whether it mildly harms them is beyond this design’s resolution, and we do not claim it.

The aggregate is blind. Overall accuracy moves +0.1 points ($b=504$, $c=508$, $p=0.925$). The reasoning gain and the knowledge shortfall offset each other. A practitioner who reads the benchmark score sees a null result and concludes the optimizer did nothing; in fact it moved accuracy between subjects at a rate their aggregate cannot express. This is the paper’s practical point, and it does not depend on any language-specific claim.

The reallocation, tested directly. A significant reasoning gain reported beside a non-significant knowledge null is two marginal tests, and the difference between them is not itself one. We therefore ran the differential estimator used throughout this paper with the label grouped by subject *type*: conditional on an item changing correctness, are the odds it changed for the worse higher for knowledge items than for reasoning items? Pooled over six models it gives 1.40 [1.08, 1.81] (bootstrap $P(\text{OR} \leq 1) = 0.008$; Table 2, last row). That is this paper’s central claim as a single estimate, and unlike the native-versus-rest contrast it survives powering: the same estimator falls from 2.67 to 1.40 and still excludes 1, where the native contrast falls from 2.27 to 1.25 and no longer does. The reallocation is the finding; the language was the coincidence.

But the models do not agree. Mantel–Haenszel assumes the strata share one odds ratio, and Cochran’s Q rejects that ($Q=14.9$, $\text{df}=5$, $p=0.011$, $I^2=66\%$): two thirds of the between-model variation is real, not sampling error. Four models fall between 1.22 and 2.42; both gemma models invert (0.59, 0.66). A DerSimonian–Laird random-effects estimate gives 1.33, and its interval depends on what one generalises over—treating the six models as exchangeable spans 1 ([0.83, 2.14]), while holding them fixed and resampling items as clusters gives [1.01, 1.73] ($P(\text{OR} \leq 1) = 0.022$). We therefore claim the reallocation *across items* and not *across*

models: established for the six we ran, not for models in general. Table 4 shows that divergence descriptively (−6.6 to +4.2 overall); this is it tested.

Powering exposes disagreement, not only shrinkage. At $n \approx 100$ the same test finds no significant heterogeneity ($Q=8.5$, $p=0.13$, $I^2=41\%$; the native contrast lower still, $I^2=20\%$, $p=0.28$). The small-sample analysis did not only overstate the effect—it made six models look as though they agreed. A placebo rotation cannot catch that: it rotates the label and leaves the pooling assumption untouched.

The reallocation begins before the optimizer. Climbing the full prompting ladder on the benchmark split (means over six models, ONA_TILI/mathematics): direct answering 32.8/53.1, CoT 33.3/74.6, BOOTSTRAPFEW-SHOT 31.0/78.5. Chain-of-thought alone buys mathematics +21.5 and the native subject +0.5; the optimizer adds a further +3.9 to mathematics and takes −2.3 from the native subject. The optimizer is not introducing the reallocation so much as extending a gradient that CoT already established—which is consistent with reasoning-subject demonstrations being the active ingredient, and with the aggregate having been uninformative for both rungs.

The correction, and what produced it. Our first pass used the benchmark’s own test split—100 ONA_TILI items per model, the scale at which subject-level audits are normally run. There, BOOTSTRAPFEW-SHOT eroded ONA_TILI in five of six models (75 CoT-correct answers flipping wrong against 46 the other way, $p=0.011$), and the comparative contrast gave an odds ratio of 2.27 [1.37, 3.91] against non-native items. We ran the obvious placebo—rotate the “native” label through every subject and re-run the identical estimator—and it passed: TARIX, the benchmark’s other knowledge subject and the closest structural analogue to a language-and-literature subject, returned 1.04 (Table 2, left).

On the frozen corpus that separation disappears. The native subject falls to 1.25 [0.98, 1.61] and TARIX rises to 1.16 [0.82, 1.64]; the gap between them goes from 1.23 to 0.09. The absolute native effect is null on every scoring (deployment $p=0.283$; knowledge-scored $p=0.602$, direction reversed; rescue-parsed $p=0.755$). Meanwhile

Table 2: The placebo rotation, before and after powering. The estimator asks: conditional on an item changing correctness, are the odds it changed for the worse higher for this subject than for the other three? At $n \approx 100$ per model the native subject separates cleanly from the other knowledge subject; on the frozen corpus at $4 \times$ the scale they converge. What survives is a reasoning/knowledge split, and the last row estimates it directly: grouping the label by subject type attenuates under powering but retains an interval excluding 1, where the native contrast does not. The two rows beneath it are the cost of that claim: the models do not share a common odds ratio ($I^2=66\%$), and an estimate that treats them as exchangeable spans 1. Powering raised I^2 from 41% to 66%—the small-sample analysis concealed the disagreement as well as inflating the effect.

Subject	$n \approx 100/\text{model}$	powered
ONA_TILI (<i>native</i>)	2.27 [1.37, 3.91]	1.25 [0.98, 1.61]
TARIX (<i>history</i>)	1.04 [0.57, 1.94]	1.16 [0.82, 1.64]
physics	0.44 [0.20, 0.86]	0.78 [0.56, 1.09]
mathematics	0.47 [0.20, 0.98]	0.75 [0.56, 1.02]
native – history gap	1.23	0.09
knowledge vs. reasoning	2.67 [1.53, 5.05]	1.40 [1.08, 1.81]
random effects	2.53 [1.17, 5.45]	1.33 [0.83, 2.14]
I^2 (models)	41%	66%

the reasoning subjects remain below 1 on both scales. The signal was never native-versus-non-native; it was knowledge-versus-reasoning, plus enough sampling noise at $n \approx 100$ to make the native subject look singular.

We note that a subject-level audit is exactly where this failure mode is invisible. Nothing about the $n \approx 100$ analysis was cutting corners: it was paired, exact, Holm-corrected, model-stratified, item-cluster-bootstrapped, and placebo-controlled. It was simply too small, and the placebo was too small along with it.

Table 3 puts every odds ratio in the paper on one axis. The four estimates above 1.6—across two languages, two serving stacks and two optimizers—all come from cells of at most 100 native items per model. The single estimate built on 393 is 1.25. We do not think this is a coincidence about which settings happen to be harmful; we think it is what a small-sample contrast looks like when it is measured repeatedly.

Per model, and what it costs to average. Table 4 breaks the audit out by model. Native accuracy falls in four of six, which on its own would be a weak claim. The informative column is the

Table 3: Every odds ratio we measured, ordered by the number of native items per model behind it. All four estimates above 1.6 come from cells of ≤ 100 ; the one built on 393 is 1.25. The pattern is not specific to a language, a stack, or an optimizer.

Setting	Optimizer	items	OR (95% CI)
TURKISHMMLU	BOOTSTRAP	65	2.35 [1.38, 4.17]
DTM split, stack A	BOOTSTRAP	100	2.27 [1.37, 3.91]
DTM split, stack B	MIPROV2	100	1.75 [0.99, 3.14]
DTM split, stack B	BOOTSTRAP	100	1.63 [0.97, 2.79]
DTM frozen	BOOTSTRAP	393	1.25 [0.98, 1.61]

Table 4: Powered audit, per model. The native subject falls in four of six models, but the divergence between the two columns is the point: aggregate movement carries almost no information about what happened to the knowledge half. gemma4:e4b is the one model where both move together, and it is also the only model with a large format-failure count (§6).

Model	Δ ONA_TILI	Δ overall
gemma4:e4b	-5.3	-6.6
qwen3.5:4b	-2.0	+4.2
qwen3.5:9b	-1.6	-0.2
qwen3.6:27b	-0.2	+1.5
qwen3.5:27b	+1.0	+0.6
gemma4:31b	+1.8	+1.1
mean	-1.1	+0.1

gap: qwen3.5:4b gains 4.2 points overall while losing 2.0 on the native subject, and qwen3.6:27b gains 1.5 overall while losing 0.2. A practitioner comparing aggregate scores across these models would rank them in an order almost unrelated to what their Uzbek-speaking users experience.

One caveat we cannot fully separate. The powered evaluation changes two things at once: more items, and *different* items (a 2019 public release rather than the benchmark). Baseline accuracies are comparable across the two (ONA_TILI 35.8 vs. ~ 33 ; TARIX 55.9 vs. 50.7), so we do not think corpus shift explains the convergence, but the design cannot rule it out and we do not claim otherwise.

5 Which Demonstrations Do Harm

If correctness-selection is the vehicle, what does it transmit? We ran the ablation that separates the candidates (Table 5): four demonstrations drawn uniformly from each model’s own correct CoT traces, three seeds per cell, split by whether they satisfy the prompt’s brevity instruction, against BOOTSTRAPFEWSHOT’s own picks.

Table 5: Demonstration ablation on ONA_TILI (means over four models; random cells averaged over three seeds). Requiring brevity-compliance is inert. BOOTSTRAPFEWSHOT’s own selection is the worst arm—below every random draw and below using no demonstrations—but that comparison rests on a single effective BOOTSTRAPFEWSHOT draw shared across models, so we report it as an observation rather than a property of the selection rule.

Arm	reasoning	payload	ONA_TILI
CoT (no demonstrations)	—	0	33.3
random, compliant	50%	1803	32.3
random, non-compliant	90%	3026	32.0
BOOTSTRAP + brevity	—	1842	31.3
BOOTSTRAP (vanilla)	—	2373	28.5
<i>constraint</i> (rand. compl. vs. non-compl.)			
$b=126, c=129$		$p = 0.900$	
<i>substitution</i> (vanilla vs. rand. non-compl.)			
$b=105, c=147$		p = 0.0097	

Not compliance. The constraint contrast is a clean null ($p=0.900$). The repair we proposed—require the demonstration to satisfy the constraint it will teach—does not work through the constraint.

Not subject mix, and not length. The random non-compliant arm averages 90% reasoning-subject demonstrations and a 3,026-character payload—more skewed and more verbose than vanilla’s 2,373—and scores 32.0, indistinguishable from the random compliant arm at 50% and 1,803. This independently reconfirms two earlier nulls of ours: a controlled experiment varying demonstration *subject* mix (four models, no contrast significant) and a 2×2 varying subject $\times$ length (long-vs-short $b=82, c=79, p=0.875$). Neither what the demonstrations are about nor how long they are determines the outcome.

An observation we cannot yet promote to a finding. BOOTSTRAPFEWSHOT’s own picks are the *worst* arm in Table 5, at 28.5: below every random draw (32.0–32.3) and below using no demonstrations at all (33.3). Taken at face value ($b=105, c=147, p=0.0097$) that would say correctness-selection picks worse-than-chance from among its own candidates—a claim about the selection rule.

It does not yet support that claim, for a reason we discovered only after running the ablation. The training pool is shuffled with a fixed seed shared across models, and BOOTSTRAPFEWSHOT keeps the first k examples that pass, so every model walks the same queue. Across six mod-

els and 24 demonstration slots the optimizer selected just **eight distinct questions**; two models selected identical sets, and most pairs overlap in two to three of four. The comparison in Table 5 is therefore between *one* BOOTSTRAPFEWSHOT draw, inherited by every model, and several independent random draws. Its pooled p -value treats four models as independent evidence when the effective number of BOOTSTRAPFEWSHOT draws is close to one. We cannot distinguish “the selection rule is bad” from “this particular draw was bad, and every model inherited it,” and we do not claim the former.

Resolving it needs a manipulation of the selection procedure rather than of the test set: recompiling BOOTSTRAPFEWSHOT under several pool orderings per model and comparing the resulting distribution against the random-draw distribution. We have not run it, and we flag that adding test items would *not* substitute for it—more items estimate the same single draw more precisely.

What the shared draw shows on its own. That eight-of-twenty-four figure is worth stating independently of the ablation. Six models spanning 4B to 31B parameters, each nominally self-selecting demonstrations tailored to itself, converge on eight questions—because selection is dominated by the ordering seed and by which items are easy enough for every model to answer, not by the model. It is also why we now read the apparent subject skew of §6 as a property of one ordering rather than of correctness-selection.

The constraint null is on firmer ground: both of its arms are random draws with three seeds each, so it is not exposed in the same way. It remains a null on 1,200 paired observations—adequate to say the constraint does not help, not to bound how small a true effect could be.

6 The Format Route Is Causal, and a Minority

One route from optimizer to failure is identifiable and repairable. The demonstrations BOOTSTRAPFEWSHOT selects are the verbose ones—selected reasoning reaches 997 characters per demonstration and 2,373 per compiled prompt—so the compiled prompt contradicts its own brevity instruction, and under a fixed budget the inflated responses truncate.

We record a correction here, because an earlier version of this analysis attributed that verbosity to

Table 6: Budget dose-response on ONA_TILI (Δ CoT $\rightarrow$ BOOTSTRAPFEWSHOT, with bootstrap-condition truncation counts). Raising the budget improves accuracy in the models that truncate and not in the control.

max_tokens	256	512	1024	2048
gemma4:e4b Δ	−10	−3	−7	−3
truncations	28	16	3	0
qwen3.5:9b Δ	−9	−5	−5	−5
truncations	13	6	3	5
gemma4:31b Δ	0	+1	0	−2
truncations	0	0	0	0

subject skew. Measured against the wrong baseline, correctness-selection appears to over-sample reasoning subjects $2.3\times$ (71% of selected demonstrations against the benchmark’s 30%). But the optimizer never sees the benchmark’s subject mix: the training pool is capped per subject and is 50% reasoning by construction. Against the pool it actually draws from, selection is 58% reasoning across demonstration slots and 50% across *distinct* selected questions—essentially no subject over-selection at all. The verbosity of the selected demonstrations is real and is what matters here; their subject composition is not the reason for it.

The budget is causal on the outcome. Truncations fall $28\rightarrow 16\rightarrow 3\rightarrow 0$ on gemma4:e4b as max_tokens rises $256\rightarrow 2048$ (Table 6), but that trend is a manipulation check—raising a cap mechanically reduces cap-hits. The causal question is accuracy. Pairing bootstrap runs item-by-item between the budget extremes, native accuracy improves in exactly the two models that truncate ($b=11$, $c=25$, $n=200$, $p=0.029$) and not in the zero-truncation control ($b=4$, $c=2$, $p=0.69$). The treated set is defined by truncation counts, not by outcome.

And it is a minority of the effect. Classifying every harmful flip on the powered corpus, 46 of 262 (17.6%) are truncation-attributable and none are format drift; the remaining 216 are well-formed, in-budget, correctly parsed answers that are simply wrong. The route is real—gemma4:e4b’s parse failures rise $72\rightarrow 223$ under vanilla bootstrapping—and it explains under a fifth of what happens.

Figure 1 shows where those failures sit: they concentrate in the two models that select the longest demonstrations, and are near-absent elsewhere. That is the scope condition on this route,

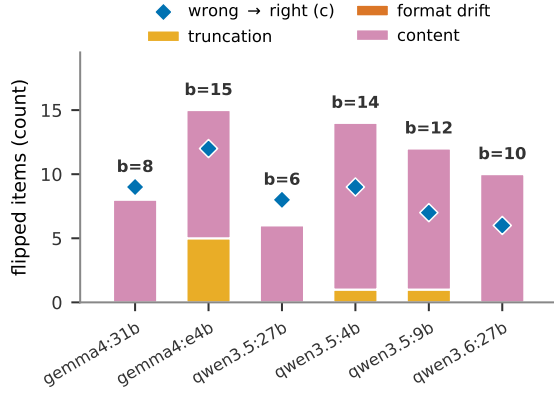


Figure 1: Harmful ONA_TILI flips classified as truncation, format-drift or content (markers: flips in the opposite direction). Format failures concentrate entirely in the models whose selected demonstration payloads are largest; everywhere else the harm is content-side and untouched by any format intervention.

and it is also why a format-side repair cannot address most of the effect.

7 Two Regimes, and Two Non-Results

Within-subject optimization is safe. Bootstrapping demonstrations from ONA_TILI training items only, evaluated across six models and $n=3,822$ paired items, finds no erosion: the benchmark half shows a small improvement ($b=129$, $c=166$, $p=0.036$ uncorrected, positive in 6/6 models) and the replication half is exactly null ($b=250$, $c=249$, $p=1.0$). The harm needs the cross-subject regime, where the optimizer can import demonstrations from elsewhere. A single-model pilot that suggested otherwise (-7.2 , $p=0.045$, $n=250$) does not survive this.

Precision is not the explanation, and cells are unstable. Serving the three large models at bf16 rather than Q4 produces erosions of -5 on two of three, so the effect is not a quantization artifact—but every model’s delta flips sign between precisions, because precision changes the correct-set and hence which demonstrations get selected. These are $n=100$ single-seed cells and we draw only the negative conclusion plus the actionable one: audits must run on the deployment configuration.

Tokenizer fertility, corrected. We hypothesised that Uzbek prose is token-expensive relative to math notation, making fixed budgets effectively smaller for ONA_TILI (cf. Petrov et al., 2023;

Lundin et al., 2025). The logged counts falsify it: math reasoning costs *more* tokens than ONA_TILI reasoning, per character (~ 0.43 vs. ~ 0.38) and per item (~ 130 – 160 vs. ~ 93 – 124). The native subject is budget-fragile because its *inputs* are longest (mean 337 characters vs. 156 for math), not because its outputs are dear.

8 Generalization

A second language. We ran the identical protocol on TURKISHMMLU (Yüksel et al., 2024) (Turkish Language and Literature against Mathematics, Physics and History; 65 test items per subject, six models). The concealment reproduces in a starker form: aggregate accuracy rises 5.2 points ($66.4 \rightarrow 71.6$) while the native subject gains nothing ($58.5 \rightarrow 58.2$)—Mathematics $+9.7$, Physics $+7.2$, History $+4.1$. The absolute native effect is null ($b=37$, $c=36$, $p=1.0$).

Turkish also shows the native-specific pattern our Uzbek data no longer supports (native 2.35 [1.38, 4.17] against History at 1.01). We do not now believe that separation without powering it: it is the same pattern, at 65 items per subject, that dissolved in Uzbek at 393. Notably History gains $+4.1$ where Uzbek TARIX does not, so the two benchmarks may genuinely differ in whether history behaves as a knowledge subject. Powering TURKISHMMLU is the obvious next experiment; until then we report the Turkish differential as unconfirmed.

A second optimizer. MIPROV2 (Opsahl-Ong et al., 2024), which searches instructions by Bayesian optimization rather than selecting demonstrations, produced the same differential magnitude as BOOTSTRAPFEWSHOT on the same items (1.75 [0.99, 3.14] vs. 1.63 [0.97, 2.79]; four treated models, both intervals including 1). Since the estimand itself attenuates under powering, we read this only as evidence that the phenomenon is not a BOOTSTRAPFEWSHOT implementation quirk.

Its mechanism evidence is sturdier and cuts the other way from payload accounts. On gemma4:e4b, MIPROV2’s demonstrations are *more* verbose than BOOTSTRAPFEWSHOT’s (mean 755 vs. 631 reasoning characters) yet produce 15 truncated and 14 unparseable generations against 62 and 56, and ONA_TILI *rises* 3 points. The difference is the instruction it wrote, demanding “a mandatory, very brief

justification” concluding with “*only* the correct single-letter choice”—it enforces the format its own demonstrations violate. Length alone does not determine the format tax; the contradiction between demonstrations and instruction does.

9 Discussion

The stable result is that automatic prompt optimization is not a uniform improvement but a *re-allocation*. It buys reasoning accuracy; it does not buy knowledge accuracy; and on our benchmark those offset each other closely enough that the aggregate metric reports nothing happened. For a national-language deployment this is the worst possible failure mode: the number used to justify shipping is exactly the number that cannot see the cost.

On our own corrections. Two claims in our first pass did not survive contact with more data: that the harm was native-language-specific, and that a brevity-constraint metric repaired it. Both were measured carefully at $n \approx 100$ —paired, exact, corrected, placebo-controlled—and both dissolved when powered. We think the generalisable lesson is not that we were careless but that subject-level audits at that scale are underpowered for the contrasts they are used to draw, and that the standard defences, placebo rotations included, are underpowered in exactly the same proportion. A placebo at $n \approx 100$ certified a separation that a placebo at $n \approx 400$ removed.

What we cannot explain. BOOTSTRAPFEW-SHOT’s demonstrations underperform random draws from the same correct pool and underperform no demonstrations at all. Subject mix, length and compliance are each measured and null. But the optimizer’s selection is also near-degenerate under a fixed pool ordering—eight distinct demonstrations across six models—so we cannot yet separate a bad selection *rule* from a bad selection. That separation is the experiment we would run next.

Four recommendations follow. (1) **Audit per subject, not in aggregate**, and on the deployment configuration. (2) **Power the audit**: our $n \approx 100$ analysis was wrong in a specific, confident and reproducible way. (3) **Randomise the pool ordering per model, and check what was selected**: with a fixed seed the optimizer returned near-identical demonstrations for models spanning an

order of magnitude in size. (4) **Score parse failures separately** from knowledge errors.

Limitations

Scale is uneven across sections. The audit (§4) rests on 5,058 held-out paired items and is the paper’s firmest result. The demonstration ablation (§5) rests on 1,200 paired observations and, for its substitution contrast, on a single effective BOOTSTRAPFEW-SHOT draw; the Turkish and MIPROV2 arms rest on $n \approx 65$ and $n \approx 100$ per model. Given that we have twice overturned our own $n \approx 100$ findings, those sections should be read as suggestive, and we say where each stands rather than presenting a uniform confidence.

Powering changes two variables. The replication corpus is both larger and different material. Baseline accuracies are comparable, but corpus shift is not fully separable from sample size here.

Greedy decoding is not reproducible for every model. Re-running the identical CoT program in a second session at temperature 0, all four qwen models reproduce item-for-item (251/251); both gemma models do not, changing correctness on 12/100 and 7/100 ONA_TILI items, for accuracy drifts of -2.0 and $+1.0$ points. That is the floor beneath every per-model gemma cell, including the dose-response deltas and the precision flips.

Scope. Two benchmarks, two languages, two optimizers, inference-time prompting only; reflective and reinforcement-style optimizers may differ. The MIPROV2 arm rests on four treated models, since the optimizer returned the baseline unchanged for two under our search budget. Subject-type grouping (knowledge/reasoning) is ours and is not the only defensible partition of four subjects.

Ethics Statement

DTM derives from publicly available national university-entrance examination materials and contains no personal data. Our aim is protective and self-correcting. We report that a widely deployed optimization method reallocates accuracy in a way its headline metric cannot express; that two of our own central claims required retraction in print after we powered them; and that our own proposed repair does not work for the reason we proposed it. All corrections appear alongside the claims they revise rather than in a footnote.

Reproducibility Statement

We release the instrumented harness, all per-item logs (~45,000 generations across ten experiments), per-experiment decision memos, the frozen replication loader, the unit-tested statistics module, and a one-command verifier that re-derives every in-text number from the logs and fails loudly on drift. Splits, option shuffling and demonstration draws are deterministic under fixed seeds; §7 and Limitations document where the serving stack is not. Code and logs: <https://github.com/idrock-ai/prompt-optimization-audit>.

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A Demonstration Composition (2×2)

The controlled manipulation behind §5’s subject and length nulls: a CoT reference plus the four cells of $\{\text{reasoning, native}\} \times \{\text{short, long}\}$, four demonstrations each drawn from the model’s own correct traces, pooled over four models on ONA_TILI, paired exact McNemar under Holm correction (Table 7). No contrast survives correction. Two cells were under-supplied—the native/long pool was empty for one model and held a single demonstration for another, because models rarely produce verbose native-language reasoning at all. Dropping both changes no verdict.

Table 7: Demonstration subject $\times$ length, pooled over four models on ONA_TILI.

Contrast	<i>b</i>	<i>c</i>	<i>p</i>	Holm
reason-short vs. CoT	51	33	.063	.378
reason-long vs. CoT	47	34	.182	.770
native-short vs. CoT	38	33	.635	1.0
native-long vs. CoT	42	29	.154	.770
long vs. short	82	79	.875	1.0
native vs. reason	78	91	.356	1.0

Table 8: ONA_TILI accuracy by arm and model; random cells averaged over three seeds.

Model	CoT	van.	compl.	r-comp	r-non
gemma4:e4b	30.0	25.0	30.0	30.7	30.3
qwen3.5:4b	33.0	28.0	31.0	30.3	30.7
qwen3.5:9b	36.0	31.0	34.0	34.3	30.7
qwen3.6:27b	34.0	30.0	30.0	33.7	36.3
mean	33.3	28.5	31.3	32.3	32.0

B Substitution Ablation, Per Model

Table 8 gives the per-model breakdown behind Table 5. BOOTSTRAPFEWSHOT’s selection is the worst arm in three of four models and tied in the fourth—though, as §5 notes, those four models largely share one selected demonstration set, so these are not four independent observations of the selection rule. On qwen3.6:27b the correctness metric and the brevity-constrained metric selected the *same* demonstrations (identical payload and identical scores), so that model is structurally uninformative about the constraint contrast rather than a measured null—the same distinction we draw for untreated MIPROV2 models.